

Jay Sejin Sim

Ph.D. Candidate, NSF GRFP Fellow

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EDUCATION

Ph.D. Student in Mechanical Engineering

September 2022 – Present

Stanford University, Stanford, CA

Advisor: Prof. Ruike Renee Zhao

Candidacy: October 2024 | GPA: 3.86/4.00

Bachelor of Science in Mechanical Engineering with Honors

May 2022

The Pennsylvania State University | Schreyer Honors College, University Park, PA

GPA: 3.85/4.00 | Dean's List: All Semesters

RESEARCH FOCUS

My research focuses on the intersection of stimuli-responsive materials and metamaterials for the development of tunable active metamaterials. Additionally, my aim is to study their performance in manipulating the propagation of elastic, acoustic, and electromagnetic waves.

HONORS AND AWARDS

Rank 3 Poster Award, Stanford Bio-X

September 2025

Future Faculty Symposium Travel Award, Society of Engineering Science

June 2025

Centennial Teaching Assistant Award, Stanford University

May 2025

Most Lab Members in Picture, Stanford Mechanical Engineering Conference

May 2024

NSF Graduate Research Fellowship, National Science Foundation

April 2024

Outstanding Undergraduate Research Achievement, PSU

April 2022

2nd place in Engineering, PSU Undergraduate Research Exhibition

April 2021

3rd place in Engineering, PSU Undergraduate Research Exhibition

April 2020

PUBLICATIONS (* denotes equally contributing authors)First-author

1. **Sim, J.**, Leanza, S., Hwang, S., Ogaki, C., & Zhao, R. R., Harnessing Twist-Enabled Instabilities for Programmable In-Plane Static and Dynamic Properties in Rod Origami (RodOri) Metamaterials. *Advanced Science* (under-preparation).
2. **Sim, J.**, Lu, L., & Zhao, R. R., Electromagnetic (EM)-Driven Functional Materials. *Advanced Materials*. e21268 (2026).
3. **Sim, J.**, Wu, S., Hwang, S., Lu, L., & Zhao, R. R., Selective Actuation Enabled Multifunctional Magneto-mechanical Metamaterial for Programming Elastic Wave Propagation. *Advanced Functional Materials*. 34(16), 2422325 (2025)
4. **Sim, J.** & Zhao, R. R., Magneto-Mechanical Metamaterials: A Perspective. *Journal of Applied Mechanics*. 91(3), 031004 (2024).
5. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R., Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation. *Advanced Materials*. 35(35), 2303541 (2023). [Selected as front cover](#).

Co-author

1. Pavlick, W.D., **Sim, J.**, West, D.L., Zhao, R. R., Ghalichechian, N., Complex Permittivity and Permeability of NdFeB Doped Polymers. *Research* (under-review).
2. Lee, D. D., Wang, L., Chen, B., Tsai, Y., **Sim, J.**, Zhao, R. R., Zhang, X. S., Chen, W., Co-Design of Programmable Material Systems. *npj Metamaterials* (2026), in-press.
3. Lee, J., **Sim, J.**, Yang, X., Wu, S., Averitt, S., & Zhao, R. R., Porous Liquid Crystal Elastomers with Thermally Tunable Acoustic Properties. *Materials Today*. 98, 103394 (2026).
4. Yang, X., **Sim, J.**, & Zhao, R. R., Spatiotemporal modulation of surface texture for information encoding and object manipulation. *Nature Communications* (2026), in-press.
5. Wu, S.*, Chang, Y.*, Leanza, S.*, **Sim, J.**, Lu, L., Stone, D., & Zhao, R. R., Magnetic Milli-Spinner for Robotic Endovascular Surgery. *Advanced Materials*. 38(4), e08180 (2025). [Selected as back cover](#).
6. Yang, X.*, **Sim, J.***, Huang, W., & Zhao, R. R., Bioinspired Synergistic Texture and Color Modulation Enabled by Surface Instability of Cholesteric Liquid Crystal Elastomers. *Science Advances*. 11(44), eaea9183 (2025).
7. Chang, Y., Li, G., **Sim, J.**, Karniadakis, G., & Zhao, R. R., Clot Treatment via Compression-and Shear-Induced Densification of Fibrin Network Microstructure: A Combined in Vitro and in Silico Investigation. *Extreme Mechanics Letters*. 102391 (2025).
8. Averitt, S., **Sim, J.**, & Zhao, R. R., AI and Computing for Active Metamaterial Design: A Perspective. *Journal of Applied Mechanics*. 92(9), 094001 (2025).
9. King, P., **Sim, J.**, Shahed, K., Harwood, C., & Manogharan, G. Hands-on digital tools for metal casting engineering education to increase student engagement in foundry science. *International Journal of Metalcasting*. 1-13 (2024).

10. Lu, L., **Sim, J.**, & Zhao, R. R., Mechanics of hard-magnetic soft materials. *Mechanics of Materials*. 189, 104874 (2024). [Selected as front cover.](#)
11. Arunachala, P. K., Abrari Vajari, S., Neuner, M., **Sim, J. S.**, Zhao, R., Linder, C., A multiscale anisotropic polymer network model coupled with phase field fracture. *International Journal for Numerical Methods in Engineering*. 125(13), e7488 (2024).
12. West, D., Pavlick, W., **Sim, J.**, Dai, J., Wu, S., Eichenberger, J., Zhao, R. R., Ghalichechian, N., Reconfiguration of Electromagnetic Metasurfaces Using Tunable Shape Morphing Structures. *2024 18th European Conference on Antennas and Propagation (EuCAP)*. (2024)
13. Martinez, D., King, P., Sama, S. R., **Sim, J.**, Toykoc, H., & Manogharan, G., Effect of freezing range on reducing casting defects through 3D sand-printed mold designs. *The International Journal of Advanced Manufacturing Technology*, 126(1-2), 569-581(2023).
14. Bate, C., King, P., **Sim, J.**, & Manogharan, G., A Novel Approach to Visualize Liquid Aluminum Flow to Advance Casting Science. *Materials*, 16(2), 756 (2023).

CONFERENCE PRESENTATIONS

1. Lee, D., Wang, L., Chen, B., Tsai, Y., **Sim, J.**, Zhao, R. R., Zhang, X. S., Chen. W. (2026) Co-Design of Programmable Material Systems: A Perspective, Poster session, Multifunctional Materials and Structures at the Gordon Research Conference, Ventura, CA, January 31, 2026.
2. **Sim, J.**, Wu, S., Hwang, S., Lu, L., & Zhao, R. R. (2025). Selective Actuation Enabled Multifunctional Magneto-Mechanical Metamaterial for Programming Elastic Wave Propagation, Oral presentation, Society of Engineering Science Annual Technical Meeting, Atlanta, GA, October 12-15, 2025.
3. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Oral presentation, Stanford Mechanical Engineering Conference, Stanford, CA, May 9, 2024.
4. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Poster session, Stanford Mechanical Engineering Conference, Stanford, CA, May 9, 2024.
5. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Magneto-Mechanical Bilayer Metamaterial with Global Area-Preserving Density Tunability for Acoustic Wave Regulation, Oral presentation, Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.
6. **Sim, J.**, Wu, S., Dai, J., & Zhao, R. R. (2023). Area-Preserving Magneto-Mechanical Bilayer Metasurface for Acoustic Wave Regulation, Poster session, Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.
7. Martinez, D., Sama, S. R., **Sim, J.**, Toykoc, H., & Manogharan, G. (2019). In-Direct Hybrid AM Metal Castings via 3D Printing, Solid Freeform Fabrication Symposium, Austin, TX, August 12-14, 2019.

SERVICE

1. Organizer, Symposium: Functional Soft Composite – Design, Mechanics, and Manufacturing, ASME IMECE, Vancouver, BC, Canada, November 8-12, 2026.
2. Stanford University Mechanical Engineering Faculty Search 2025-2026 Student Committee, February – March 2026.
3. Session chair, Symposium: 6.1 Functional Soft Composites – Design, Mechanics, and Manufacturing. Society of Engineering Science Annual Technical Meeting, Minneapolis, MN, October 8-11, 2023.

TEACHING

Course Assistant and Lab Instructor

ME 303: Soft Composites and Soft Robotics (~120 students per year)

Fall 2023, 2024, 2025, Spring 2025

Department of Mechanical Engineering, Stanford University

- Designed lab sessions (3 total per quarter) and taught students experimental fabrication and characterization of soft materials, soft electronic sensors, and shape-morphing structures.
- Held weekly office hours, lab sessions, graded assignments, and gave ABAQUS finite element analysis lecture.
- Class size: 55-65 students. [Final projects](#).
- Teaching evaluations available upon request.

OUTREACH

Stanford seeME Instructor

Winter - Spring 2026

Stanford Department of Mechanical Engineering, Stanford University

- Designed a class on origami applications in mechanical engineering for local high school and middle school students at an annual on-campus outreach event.

Summer Undergraduate Research Fellowship (SURF) Mentor

Summers 2024, 2025

Stanford College of Engineering, Stanford University

The program has a strong focus on training underrepresented minority or first-generation undergraduate student engineers who can promote diversity in STEM.

- Mentored an undergraduate student from Tufts University on the basics of experimental research and technical communication. The student was trained to conduct research and present an end of program poster session.
- Stimulated the student's interest in pursuing a Ph.D. in mechanical engineering.

NSF REU Mentor

Summers 2023 – 2026

Stanford Undergraduate Research Institute (SURI), Stanford University

- Mentored and trained multiple Stanford University mechanical engineering undergraduate students in general mechanical engineering research practices and helped prepare students for the end of program poster session.
- Fulfilled this role across multiple summer academic quarters at Stanford University.

Summer Research Program Mentor

June 2023 – July 2023

Stanford Robotics Center, Stanford University

- Mentored two local Bay Area high school students in STEM concepts in a six-week program designed to promote robotics and engineering education in first generation, minoritized, and underrepresented students.

Undergraduate Research Ambassador

August 2020 – May 2022

Undergraduate Research and Fellowships Mentoring Office, The Pennsylvania State University

- Held weekly peer-advising hours for other undergraduate students wanting to be involved in undergraduate research.
- Presented undergraduate research opportunities on campus to classes of first-year engineering students.

MENTORING

Munashe Chibinyani	Summer/Fall 2026	Ph.D. Rotation, Mechanical Engineering, Stanford University
Kendra Chang	2026	BS Mechanical Engineering, Stanford University
Yizhi Chen	Summer 2026	BS Mechanical Engineering, University of Michigan
Yufei Chen	2026	BS Computer Science, Stanford University
Joseph Jimenez	Spring 2026	BS Mechanical Engineering, Stanford University
Ellia Cavanaugh	Spring 2026	BS Mechanical Engineering, Stanford University
Chloe Kim	2025 – 2026	BS Mechanical Engineering, Stanford University
Chisa Ogaki	2025 – 2026	BS Mechanical Engineering, Stanford University
Sarah Hwang	2023 – 2026	BS/MS Mechanical Engineering, Stanford University
Samantha Averitt	Fall 2025	Ph.D. Rotation, Mechanical Engineering, Stanford University
Ariana Olivares	2024 – 2025	BS Mechanical Engineering, Tufts University
Caleb Nangle	Winter 2025	BS Mechanical Engineering, Stanford University
Phoenix Clarke	Summer 2024	BS Mechanical Engineering, Stanford University
Annika Yong	2023 – 2024	MS Mechanical Engineering, Stanford University
Eli Jones	Spring 2023	BS Mechanical Engineering, Stanford University

3 high school students mentored, names omitted for privacy